

Final Exam Study Guide

MTH 310, Spring 2026

Format and Length

- **Cumulative** — covers Chapters 12–16.
- **Approximately 11 problems**, mostly 18 points each, with one or two slightly longer problems worth a bit more. Total: **200 points**.
- **Two hours**. You may use a calculator (rarely useful). Show all work for credit.
- Format mirrors the in-class exams: free response, with workspace.

Topic Distribution

Based on last year's final, the points were spread roughly as follows. Expect a similar weighting:

Chapter	Approximate % of points	Approximate # of problems
Chapter 12 (Vectors, lines, planes)	~10%	1
Chapter 13 (Vector functions, curves)	0–5%	0–1
Chapter 14 (Partial derivatives, optimization)	~35%	4
Chapter 15 (Multiple integrals)	~10%	1
Chapter 16 (Vector calculus)	~45%	5

Bottom line: Chapter 14 and Chapter 16 dominate. Together they make up about 80% of the exam. Chapter 12 will likely show up briefly, and you can expect at least one multiple-integral problem from Chapter 15.

What to Prioritize (in rough order of importance)

High priority — nearly certain to appear

- **14.7 Optimization on a closed region.** Find interior critical points (set $\nabla f = 0$, classify with the second-derivative test if asked), then check the boundary — often via parametrization. Compare values. Lagrange multipliers (14.8) is fair game too.
- **14.6 Gradient and directional derivative.** Direction of fastest increase/decrease, rate of change in a given direction.
- **14.4 Tangent planes and linearization.** Both for graphs $z = f(x, y)$ and for level surfaces $F(x, y, z) = 0$.
- **14.2 Limits.** “Does the limit exist?” — show 0 along all lines of one form, then find a path that gives a different value (a parabola is a classic).
- **15.3, 15.7, 15.8 Multiple integrals in polar/cylindrical/spherical.** Choose the right coordinate system; set up bounds; evaluate.
- **16.2 Line integrals.** Both scalar $\int_C f \, ds$ and vector $\int_C \mathbf{F} \cdot d\mathbf{r}$. Practice parametrizing standard curves: line segments, parabolas, helices.

- **16.3 / 16.5 Conservative fields and FTLI.** Curl test, find a potential, apply FTLI.
- **16.4 Green's Theorem.** Recognize when it's the right tool (closed curve in plane, especially with hard-to-integrate P or Q). The area formula $\frac{1}{2} \oint (x dy - y dx)$ is fair game.
- **16.6 / 16.7 Surface area and surface integrals.** Be comfortable with both setups: graph $z = g(x, y)$ and parametric (u, v) . Know the spherical parametrization cold.
- **16.8, 16.9 Stokes' and Divergence Theorems.** Recognize the appropriate situation for each.

Medium priority — might appear

- **12.5 Lines and planes.** Equation of a plane through three points, line through a point parallel/perpendicular to something, intersection of a line and a plane.
- **14.3 Partial derivatives.** Higher-order derivatives, mixed partials.
- **14.5 Chain rule.** Standard tree-diagram chain rule for compositions.
- **15.1–15.2 Iterated double integrals.** Including reversing the order of integration.
- **13.1–13.2 Vector functions.** Parametrize a curve given as the intersection of two surfaces; tangent vectors.

What You Can De-prioritize

These topics either didn't appear last year or are unusually narrow; don't skip them entirely, but they're not where to spend your last hour:

- **Curvature, principal normal, binormal (13.3–13.4).** Arc length is fair game in a generic line-integral; the TNB framework itself is unlikely.
- **Differentials (14.4)** as a standalone topic. The tangent-plane interpretation is what matters.
- **Implicit differentiation in chain-rule form** ($\partial z / \partial x$ from $F(x, y, z) = 0$). Useful, but not central.
- **12.4 Cross product computations as their own topic.** Cross products are a tool everywhere in Chapter 16, but a standalone “compute $\vec{u} \times \vec{v}$ ” problem is unlikely.
- **Möbius strips and non-orientable surfaces.** Conceptually interesting but a poor exam target.
- **Generalized Stokes' Theorem / differential forms.** Mentioned for closure; not testable.
- **Heat-flow physical applications** of the Divergence Theorem. The math is testable; the physics scenario is unlikely.
- **Vortex-field / non-simply-connected hypothesis** discussions. Important conceptually — expect this only if it shows up as a brief T/F or one-sentence justification, not as a 20-point problem.

Two-Step Study Plan

Step 1: build the formula tables. The biggest source of confusion in Chapter 16 is the proliferation of “ dS for graph form” versus “ dS for parametric form”. Make yourself a single one-page comparison: graph $z = g(x, y)$ on the left, parametric (u, v) on the right, with dS , area, and flux. Use the Chapter 16 review handout as a starting point.

Step 2: re-do quizzes 6, 7, 8, 9, and Test 3. Those collectively cover the bulk of Chapters 14–16. The exams in particular show what kinds of setups and computations have been emphasized.

Practice Materials

- **Released quizzes 1–9 with solutions** on the course website.
- **Tests 1, 2, 3 with solutions** on the course website.
- **Practice problems 1–2** on the course website.
- Re-read the **Chapter 16 review handout** (separate document) for the formula tables and comparisons.